

Ensemble Learning

Wisdom of the crowd

Ensemble learning is a machine learning paradigm that combines multiple individual models (often called "base learners" or "weak learners") to produce a single, stronger and more robust predictive model.

It improves accuracy and reduces overfitting by aggregating predictions - usually through voting or averaging - to mitigate the errors of any single model. Common techniques include bagging (random forest), boosting and stacking.

stacking

Types of Ensemble Learning

1) Voting ensemble - It aggregates predictions via hard voting (majority rule) or soft voting (avg probabilities) to reduce variance and mitigate

2) Bagging (Bootstrap Aggregation)
Trains multiple models in parallel on random subsets of the data and aggregate their prediction (e.g. Random Forest). This reduces variance and overfitting

3) Boosting → Ada Boosting, Gradient, XGB
Trains models sequentially, where each new model attempts to correct the

errors of the previous ones. This reduces bias and variance.

4) Stacking : Trains multiple, often different, models (base learner) and uses a meta-model to combine their predictions.

Benefits

- 1) Improvement in performance
- 2) Bias & Variance get reduced.
- 3) Robustness

When to use?

Always.